

Phase 1 Substrate Extras

PREPARED BY Energy Hippo implementation team

Three small but report-blocking additions that finish Phase 1: **bill status history**, a few **digital/status points**, and sample **notes & images**. These are additions to `data/scenarios/demo-university-v1.yaml` (channels/points) and the `data/eem/` setup (status history, notes, images), authored here as load-ready spec.

1. Bill status history (StatusHistory)

Why. `rp_LateBills`, `rp_DailyProcessingSummary`, and `rp_FiscalPeriodBills` derive entry dates and processing users from `StatusHistory` (Table_Code BL). Without it those reports are empty and "late" can't be computed.

Rule. Every loaded bill gets at least a `Status_Code` 1 (Bill Entry) history row:

field	value
Table_Code	BL
Status_Code	1 (Bill Entry); 2 (QC Approval) for the approved subset
SH_TableRowID	the bill id
AsOfDate	<code>billing_end</code> + entry lag (see below)
LastChangedBy	the entry/import user

Entry lag + users (deterministic):

Provider	Entry path	LastChangedBy	Entry lag after period end
Valley Electric	import	nora.chen	3 days
Sierra Gas	manual entry	iris.morales	5 days
River City	import	nora.chen	4 days
Northlake Thermal	manual (allocation)	keiko.tan	6 days
Helios Solar	manual entry	iris.morales	7 days
Cedar Row (VED/SGU/RCU/HOS)	manual entry	marcus.reed	5 days

This yields ≥ 2 distinct entry users overall (requirement) and realistic per-provider lags so `rp_LateBills` (default 5-day overdue) flags the slow ones.

Late-bill fixture. Stop one account's series one period early (no current bill) so its projected next-entry date is > 5 days overdue → it appears in `rp_LateBills`. Use **Sierra Gas – Library** (low-traffic) for the deliberate late account; document it as a scenario event `late_bill_fixture`.

Approved subset. Mark ~1/3 of bills Status_Code 2 (QC Approval) by an AccessLevel-3 user (per data/eem/northlake-eem-permissions-v1.yaml) so AP-export eligibility (Phase 3) and rp_DailyProcessingSummary approval counts are non-trivial.

2. Digital / status points

Why. `rp_DigitalSummary` and the digital side of the interval engine (`DigitalArchive_TBL`) need points whose `point_type` is `Digital` with an enumerated `state_set` (see `digital_reading` in `implementation/synthetic-source-model.pdf`). Northlake's only Digital point so far is Chiller Enable Status on the Central Plant Chiller Controller, and its input path is still planned.

Add these channels / points (IDs follow existing manifest conventions; the chiller channel already exists in the scenario):

channel id	source	commodity	point_type	state_set	interval	maps to
<code>ch_modbus_ntp_chiller_enable_status</code>	Central Plant Chiller Controller (existing Modbus meter; point Chiller Enable Status)	electric	Digital	<code>chiller_run {0 off, 1 on}</code>	5 min	<code>DigitalArchive_TBL</code>
<code>ch_student_center_occupancy_status</code>	Student Center BMS	control	Digital	<code>occupancy {0 unoccupied, 1 occupied}</code>	15 min	<code>DigitalArchive_TBL</code>
<code>ch_science_center_lab_exhaust_status</code>	Science Center BMS	control	Digital	<code>exhaust_mode {0 normal, 1 high}</code>	15 min	<code>DigitalArchive_TBL</code>

Reading shape (`digital_reading`). `state_value` is from the set (never a measured quantity). Deterministic patterns:

- `chiller_run`: on during cooling season daytime, off overnight/winter.
- `occupancy`: occupied Mon–Sat building hours, unoccupied otherwise; extra occupied blocks on Student Center event days.
- `exhaust_mode`: mostly normal, high during weekday lab hours.

These also give `rp_DigitalSummary` runtime/duty-cycle results with known totals (e.g. chiller on-hours per month).

3. Notes & images

Why. `rp_NoteHistory` and `rp_ImageManagement` need `NoteHistory_Tbl` / `Images` rows attached to entities. Bill images also support the bill-image viewer used across bill reports.

Notes (note entity → `NoteHistory_Tbl`):

target	table code	note
Library electric meter (replacement)	METR	"Meter replaced 2024-08-14; new synthetic serial issued, same account; reads continuous."
Sierra Gas Student Center estimated bill	BL	"Estimated read this cycle; corrected actual issued next cycle (estimated_bill_correction)."
Cedar Row A RCU account (renewal)	BA	"Account number changed at the July 2024 renewal; site/building continuity preserved across its water and sewer services."
Valley Electric Science Center	METR	"Lab load drives summer peak; power-factor adjustment expected on heavy months."

Images (image entity → `Images`):

target	table code	image_kind	file_ref
Each provider's sample bills	BL	bill_image	the generated PDFs under documents/utility-bills/<provider>/...
Library replacement meter	METR	meter_photo	synthetic placeholder photo

The bill-image links reuse the already-generated synthetic PDFs, so the bill-image path is exercised with real artifacts.

expectedAssertions

- `every_bill_has_entry_history` — each loaded bill has a `Status_Code 1 StatusHistory` row with a valid `AsOfDate` and `LastChangedBy`.
- `distinct_entry_users` — ≥ 2 distinct `LastChangedBy` across bill entry.
- `late_bill_detected` — the Sierra Gas Library late fixture appears in `rp_LateBills`.
- `digital_points_present` — ≥ 3 Digital points with `DigitalArchive` data; `rp_DigitalSummary` returns on-hours per month per point.
- `notes_and_images_attached` — `rp_NoteHistory` and `rp_ImageManagement` return the seeded notes and bill images.